Contents

Syllabus

Data Visualization Module

Lecture 1: Introduction to the course

Introduction to the Data Visualization module; textbooks; definitions and terminology; visual perception; pre-attentive attributes; Gestalt principles

[**slides**]

reading material: Chapter 1 [@munzner2014visualization]

Lecture 2: Nested model

Analysis framework: nested model; data abstraction (what); common types of data; task abstraction (why)

[**slides**]

reading material: Chapter 2,3,4 [@munzner2014visualization]

resources: Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 2]; [video Chapter 3]; [video Chapter 4]

Lecture 3: Visual encoding

Visual encoding; marks and channels; color in visualization; color palette; color deficiency; color spaces

[slides]

reading material: Chapter 5: Marks and Channels, Chapter 10: Map Color and Other Channels [@munzner2014visualization]

resources: VizPalette **resources:** Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 5-a]; [video Chapter 5-b] [video Chapter 10-a]; [video Chapter 10-b]; [video Chapter 10-c]

Lecture 4: Common charts

Visualize tabular data; common visual idioms and charts; scatterplot; (stacked) bar chart; streamgraph; dot/line chart; Gantt chart; slopegraph; heatmap; radial bar chart; star plot; radar plot; pie chart; coxcomb chart; parallel coordinates; dual-axis charts; Visual vocabulary

[slides]

reading material: Chapter 7: Arrange Tables [@munzner2014visualization]

resources: Visual Vocabulary **resources:** Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 7-a]; [video Chapter 7-b]

Lecture 5: Static plotting in Python

Static plotting in Python; basic plotting in matplotlib; style and personalize plots; towards more advanced plotting in matplotlib; introduction to seaborn; rules of thumb in data visualization

[slides-a] [slides-b]

tutorials: - `01-matplotlib/01-figures_subplots.ipynb` — figures and subplots - `01-matplotlib/02-plotting.ipynb` — basic plotting - `01-matplotlib/03-customizing_appearance.ipynb` — style and customisation - `01-matplotlib/04-visual-encoding.ipynb` — visual encoding: marks, channels, and color - `01-matplotlib/05-advanced_subplotting.ipynb` — advanced subplotting - `01-matplotlib/06-charts-matplotlib.ipynb` — common charts (scatter, bar, line, histogram) with Gapminder

resources:

- matplotlib: Examples | Tutorial | User guide
- seaborn: Gallery | Tutorial | API

Lecture 6: Geographical plotting in Python

The importance of spatial thinking; exploring spatial phenomena visually; how to create informative thematic cartography; main thematic cartography types and rules of thumbs; pitfalls of using spatial data in computational disciplines

[**slides**]

Lecture 7: Visualize static networks in Python

Visualize static networks using NetworkX and Matplotlib

Lecture 8: Interactive Views

How to handle complexity; manipulate: change, select, navigate; facet: juxtapose, partition, superimpose; reduce: filter, aggregate, embed

[**slides**]

tutorials: - `02-interactive/01-plotly-express.ipynb` — interactive charts with Plotly Express (same 4 charts + animated bubble chart) - `02-interactive/02-altair.ipynb` — Grammar of Graphics with Altair (linked selections, layering)

resources: Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 11-12a] [video Chapter 11-12b] [video Chapter 13] [video Chapter 14]

Lecture 9: Introduction to D3.js

Why D3, why now: when declarative grammars run out; the web platform in 10 minutes (HTML, CSS, JavaScript, the DOM, SVG); SVG attributes as visual channels; D3 in three ideas (selections, scales, joins); the margin convention; building the Gapminder scatterplot end-to-end; Rosetta stone — the same chart in matplotlib, Altair, and D3

[**slides**]

tutorials: - `09-d3js-intro/09a-building-svg-with-d3.html` — HTML scaffolding, `select/append/attr/style`, SVG element catalog, coordinate system, attribute playground - `09-d3js-intro/09b-data-driven-svg.html` — the data join (`selectAll · data · join`); explicit enter / update / exit decomposition - `09-d3js-intro/09c-scales-and-axes.html` — scales (linear / log / band toggle), margin convention, axes; same skeleton on Gapminder - `09-d3js-intro/09d-loading-data.html` — `d3.csv` accessor; helpers (`extent`, `group`, `rollup`); formatters (`d3.format`, `d3.timeFormat`) - `09-d3js-intro/09e-first-scatterplot.html` — capstone: full Gapminder scatterplot end-to-end

reading material: - Murray, *Interactive Data Visualization for the Web* (2nd ed.), chapters 1–6 — free online - Bostock, "Thinking with Joins"

resources: d3js.org | Observable D3 collection

> Legacy slide decks `dv09-D3-course-lesson-1.pdf` and `dv09-D3-course-lesson-2.pdf` (third-party material) are kept under `slides/` as supplementary reference.

Lecture 10: Advanced D3 — Update Pattern, Transitions, Interactions, Layouts, Zoom

The five moves that turn a static D3 chart into a living one: **change** (the update pattern with `selection.join` and key functions), **time** (transitions: duration, delay, easing, chained, staggered, shared timeline), **input** (event listeners and the event → state → render loop; hover, click-toggle, slider, brush), **composition** (layouts: bar chart with `scaleBand`, stacked bar with `d3.stack`, streamgraph with offsets and curves), **navigation** (`d3.zoom` rescale-the-scales pattern with clip path)

[**slides**]

tutorials: (index) - `10-d3js-shapes-layouts/10a-update-pattern.html` — General Update Pattern; `selection.join`; key functions vs index joins; custom enter/update/exit callbacks; sortable bar chart - `10-d3js-shapes-layouts/10b-transitions.html` — transitions, easing, chaining, staggered delays, enter/exit transitions, bar chart race - `10-d3js-shapes-layouts/10c-interactions-capstone.html` — event → state → render loop; hover, click-toggle, year slider, brush; interactive Gapminder capstone - `10-d3js-shapes-layouts/10d-examples-gallery.html` — layout-driven charts: bar chart, `scaleBand` padding, stacked bar with `d3.stack`, streamgraph; *which layout for which question?* decision table - `10-d3js-shapes-layouts/10e-zoom-and-navigation.html` — `d3.zoom` behaviour; rescale-the-scales pattern; clip path; production checklist

examples: 24 self-contained HTML demos in `examples/` (linked from the index above), one per concept.

> Legacy slide decks `dv10-D3-course-lesson-3.pdf` and `dv10-D3-course-lesson-4.pdf` (third-party material) are kept under `slides/` as supplementary reference.

Lecture 11: Networks with D3 — NetworkX→D3, Force Layouts, Interactivity, Encoding, Performance

From the NetworkX/D3 JSON interchange to a complete interactive network viewer: **data** (`nx.node_link_data` and `d3.json`, the nodes/links contract, source/target string-then-object lifecycle), **layout** (`d3.forceSimulation` with `forceLink`, `forceManyBody`, `forceCenter`, `forceCollide`; tuning strength and distance; pinning nodes), **interactivity** (drag, hover-neighbour highlighting, click-to-pin, click-to-isolate), **encoding** (size, color, community hulls; zoomaware labels), **scaling** (progressive build, performance budget, web workers for off-main-thread simulation).

tutorials: (index)

- `11-networks-d3js/11a-networkx-to-d3.html` — node-link JSON interchange; nodes/links contract; fixed circle and random-layout baselines
- `11-networks-d3js/11b-force-layouts.html` — `d3.forceSimulation`; link/charge/center/collide forces; tuning; pinning nodes
- `11-networks-d3js/11c-interactivity.html` — drag behaviour; hover-neighbour highlighting; click-to-pin; click-to-isolate
- `11-networks-d3js/11d-encoding-and-zoom.html` — visual encoding for graphs; community overlays via convex hulls
- `11-networks-d3js/11e-capstone.html` — capstone: Game of Thrones network viewer combining 11a–11d
- `11-networks-d3js/11f-progressive-build.html` — same viewer rebuilt incrementally with performance and scale notes
- `11-networks-d3js/11h-web-workers.html` — moving `d3.forceSimulation` off the main thread with a Web Worker

examples: 21 self-contained HTML demos in `examples/` (linked from the index above), one per concept.

> Legacy slide deck `dv11-D3-course-lesson-5.pdf` (third-party material) is kept under `slides/` as supplementary reference.

Lecture 12: Graphs Visualization

Interactive graph visualization; advanced examples

[**tutorial**]

References